A Model Selection Approach to Interference

Appendix

Gustavo Diaz*

May 14, 2021

A. Extended Simulation Results

Figures A1-A4 show simulations results with $\tau = 0.26$. Figures A5-A8 do so for $\tau = 0.63$ The interpretation of the simulation results is the same as in the main text.

*PhD Candidate. Department of Political Science. University of Illinois at Urbana-Champaign. E-mail: diazdia2@illinois.edu

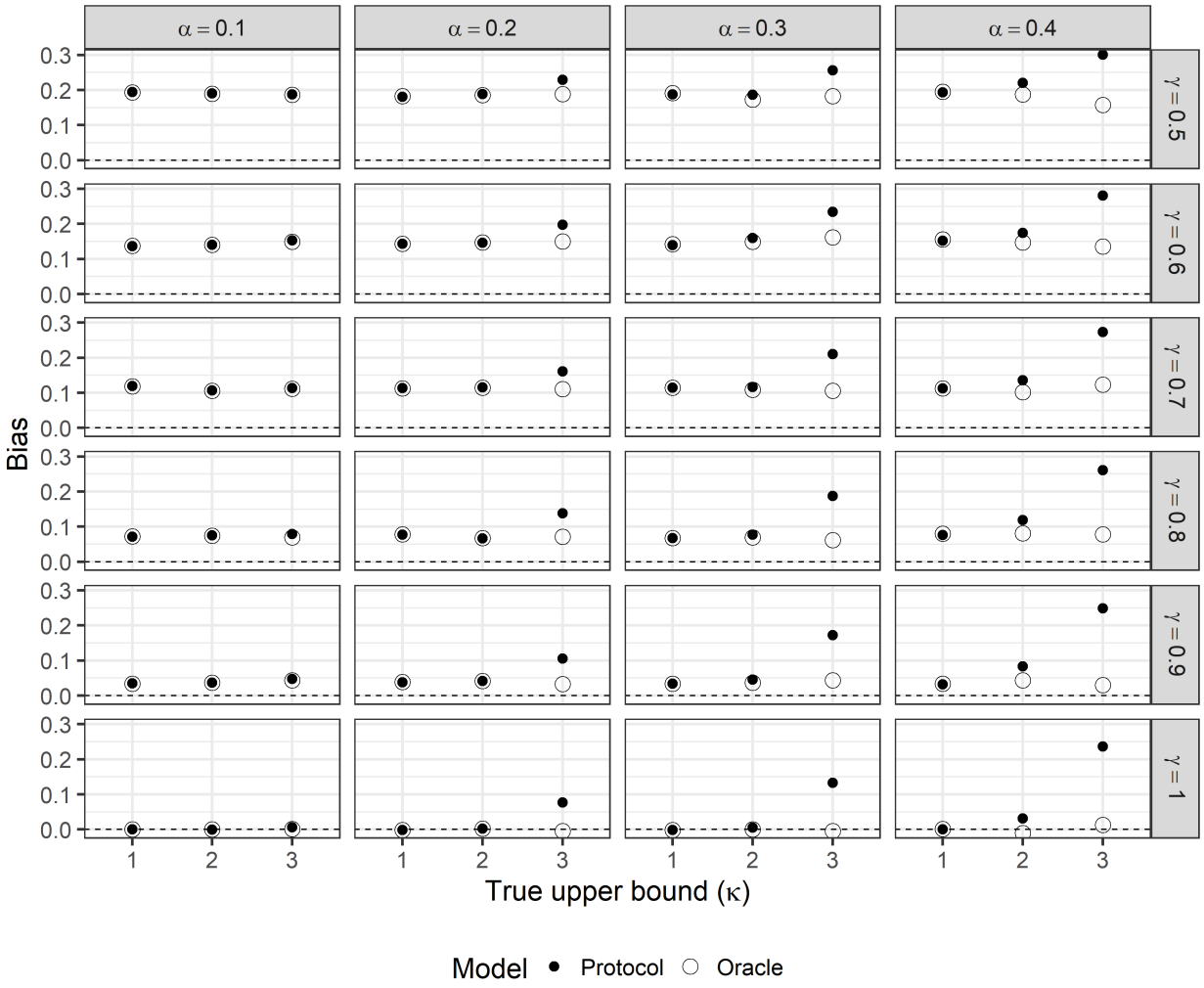

Figure A1: Bias for simulations with $\tau = 0.26$

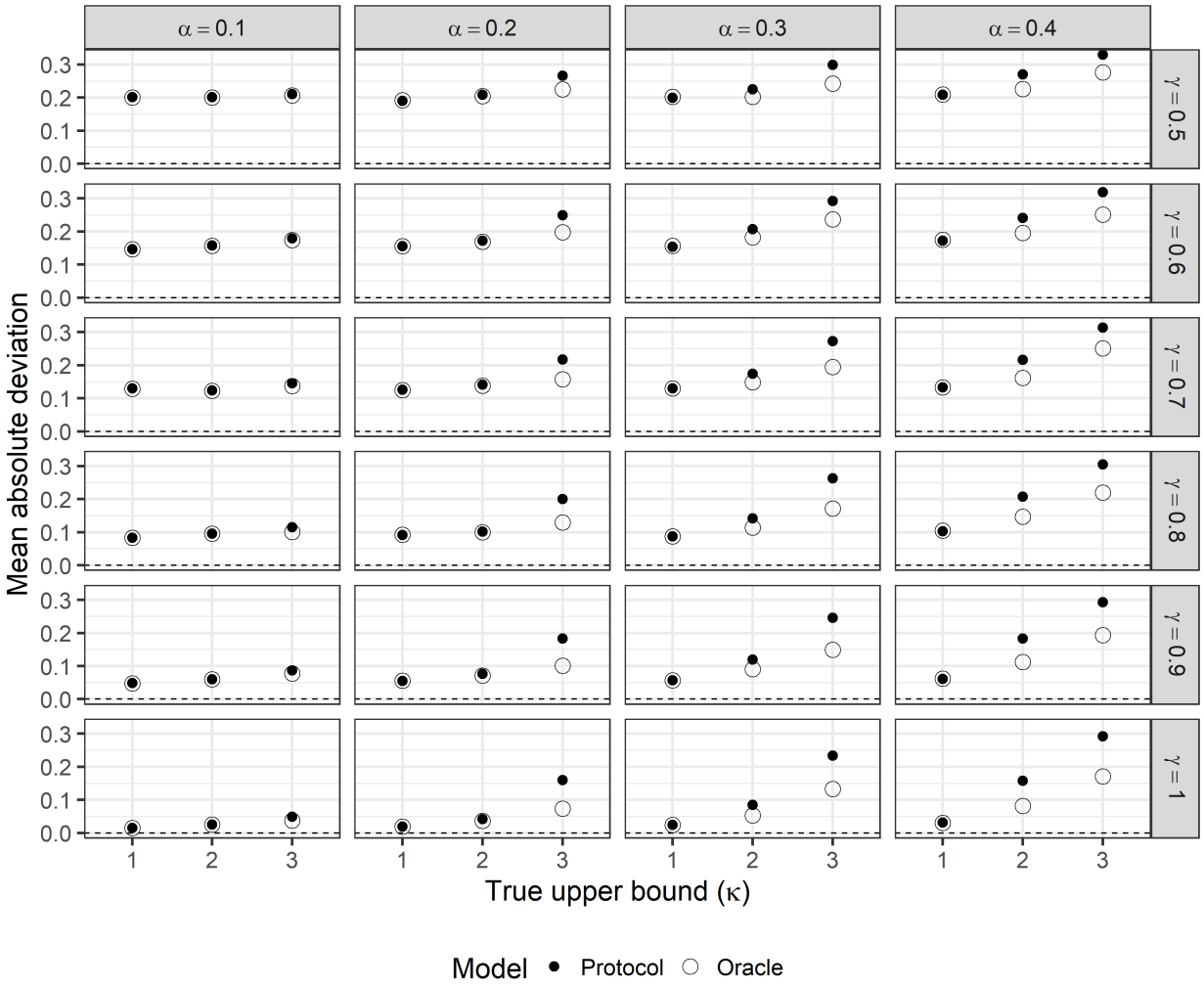

Figure A2: Mean absolute deviation for simulations with $\tau = 0.26$

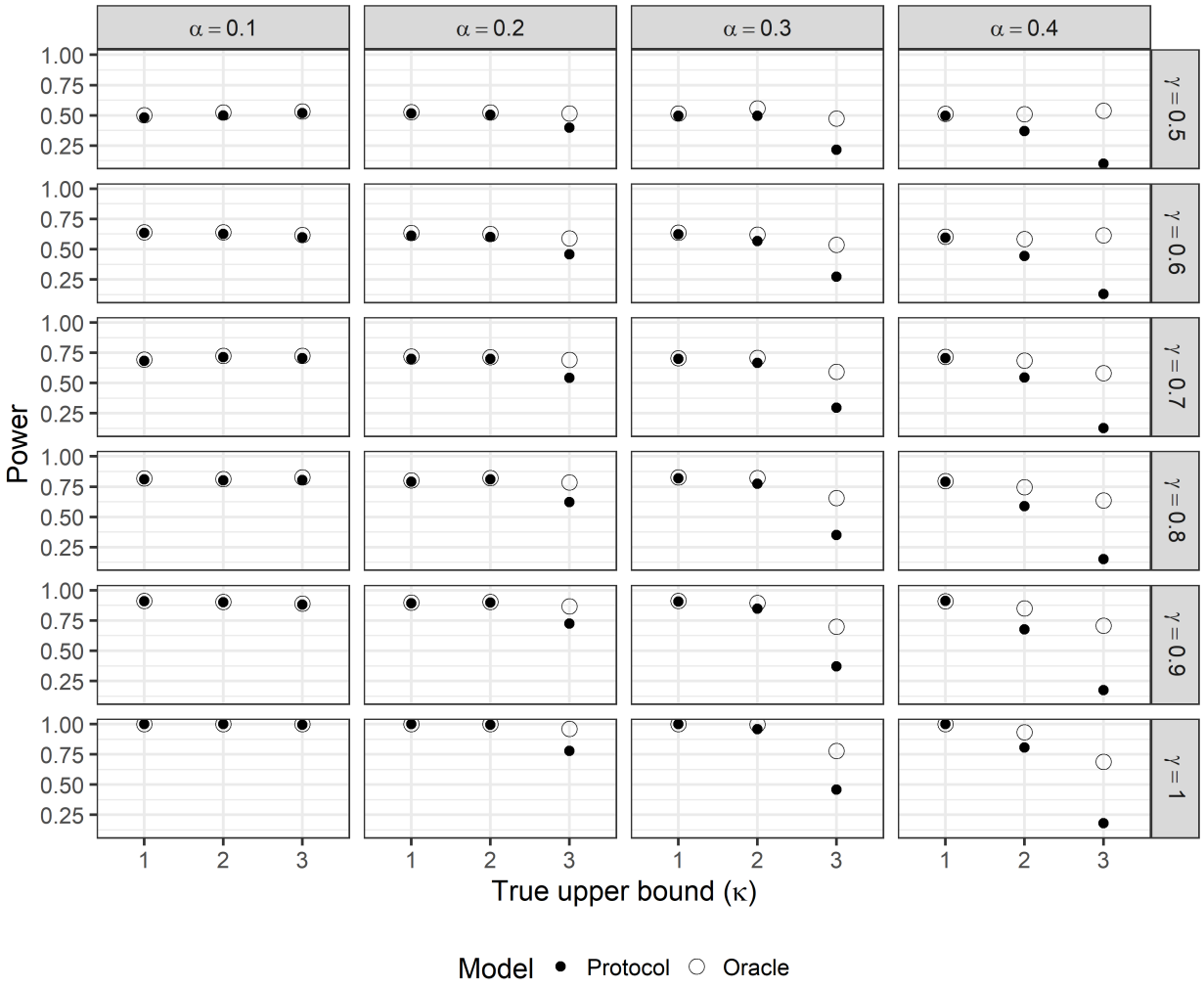

Figure A3: Power for simulations with $\tau = 0.26$

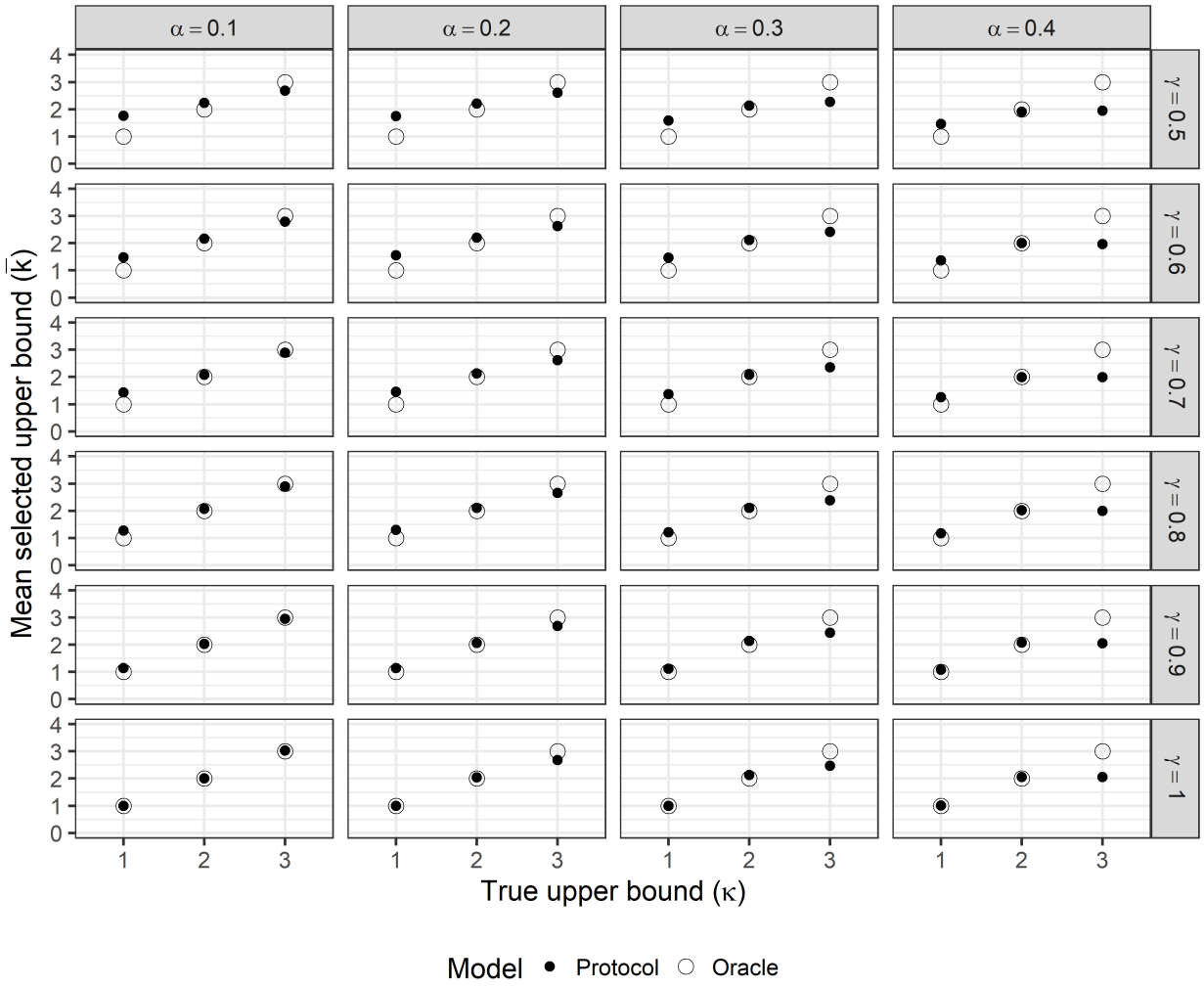

Figure A4: Mean selected upper bound for simulations with $\tau = 0.26$

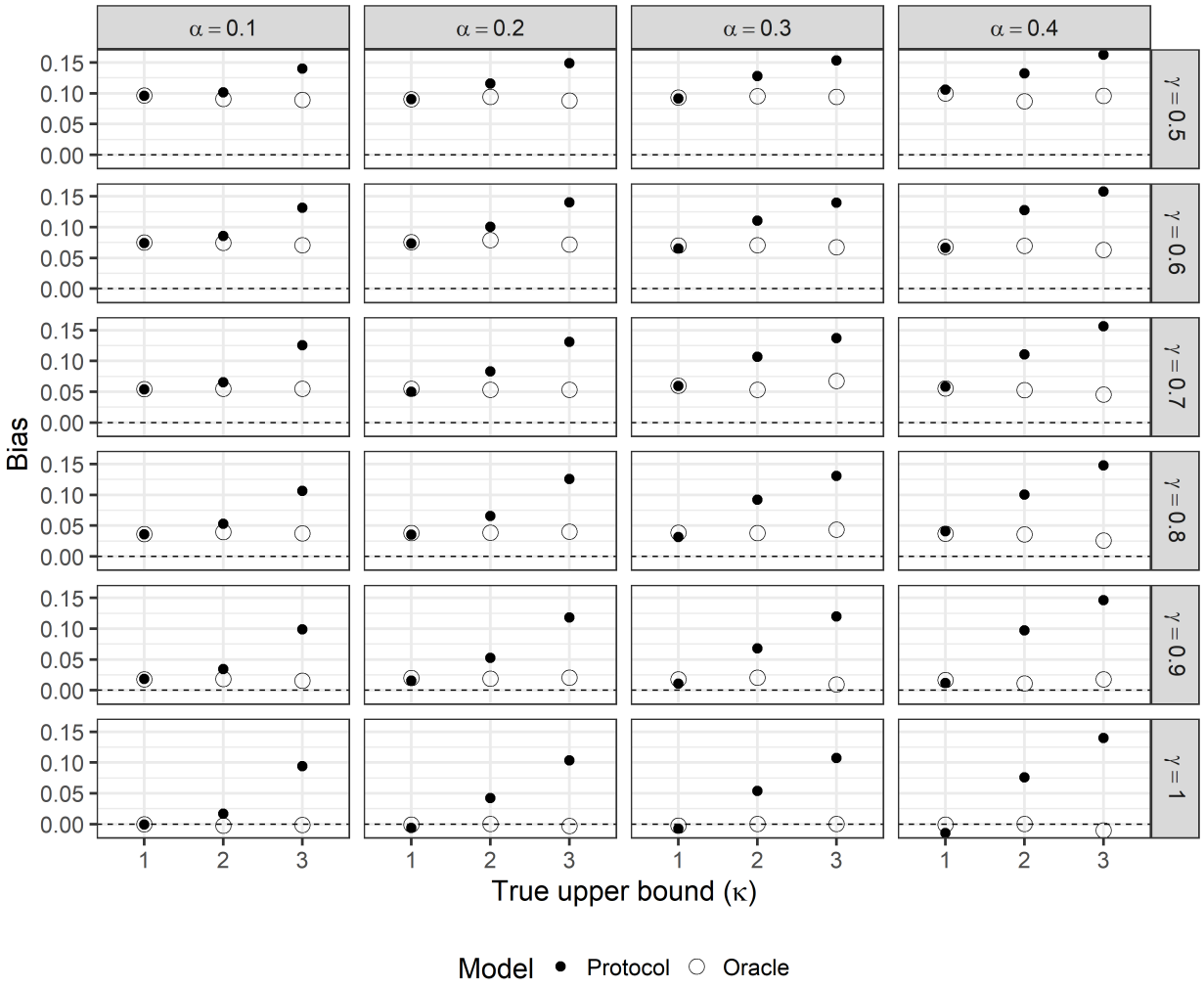

Figure A5: Bias for simulations with $\tau = 0.63$

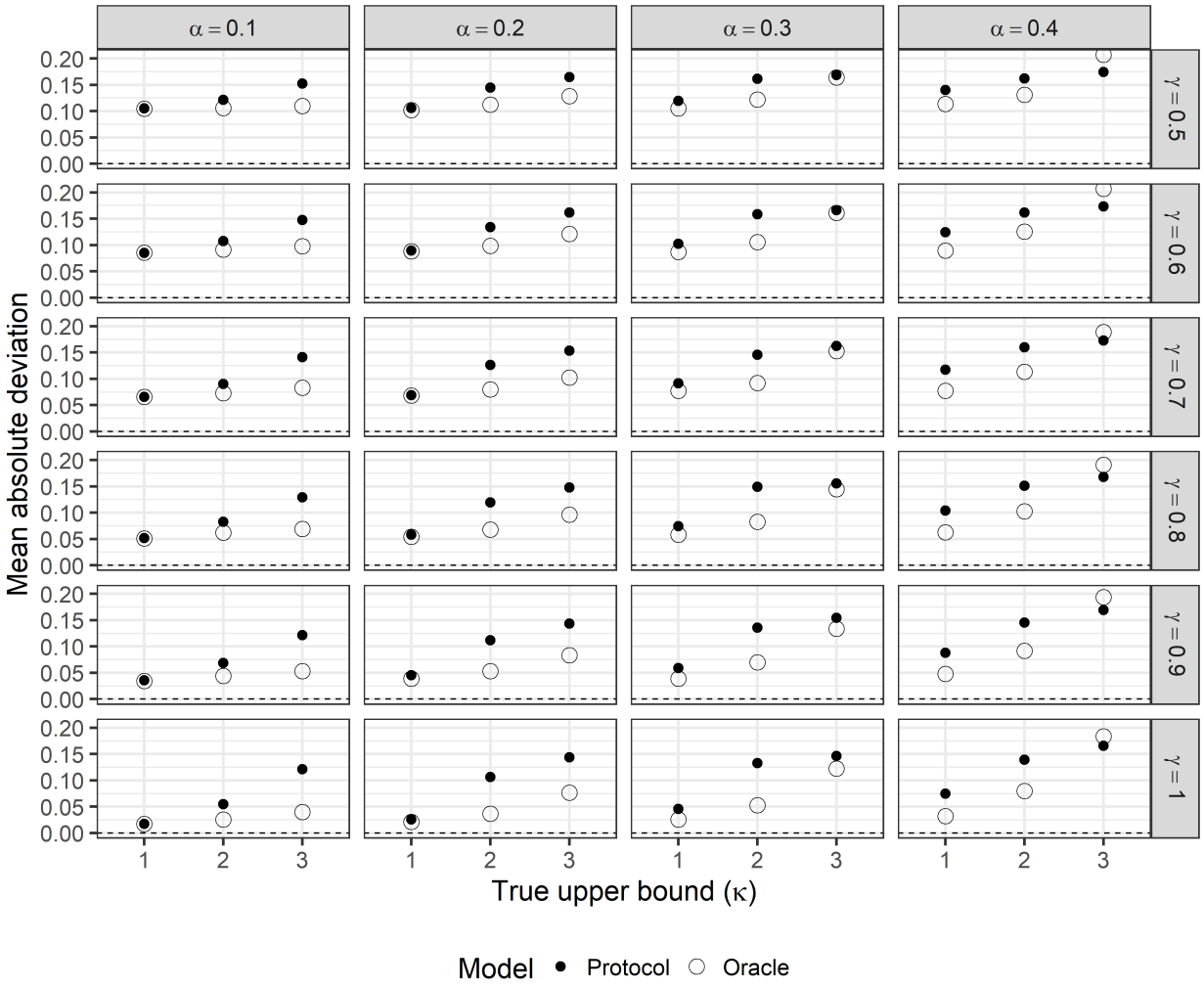

Figure A6: Mean absolute deviation for simulations with $\tau = 0.63$

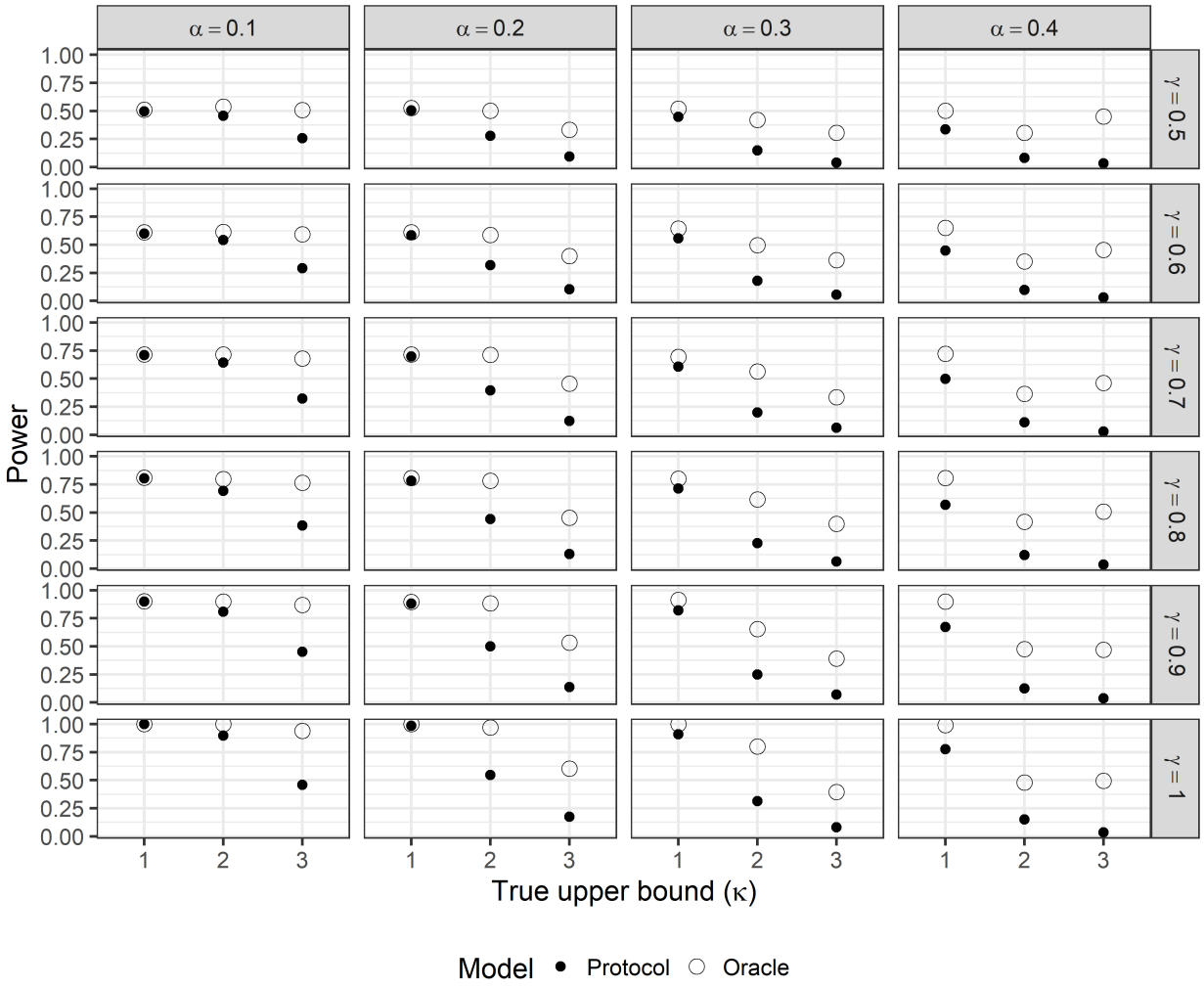

Figure A7: Power for simulations with $\tau = 0.63$

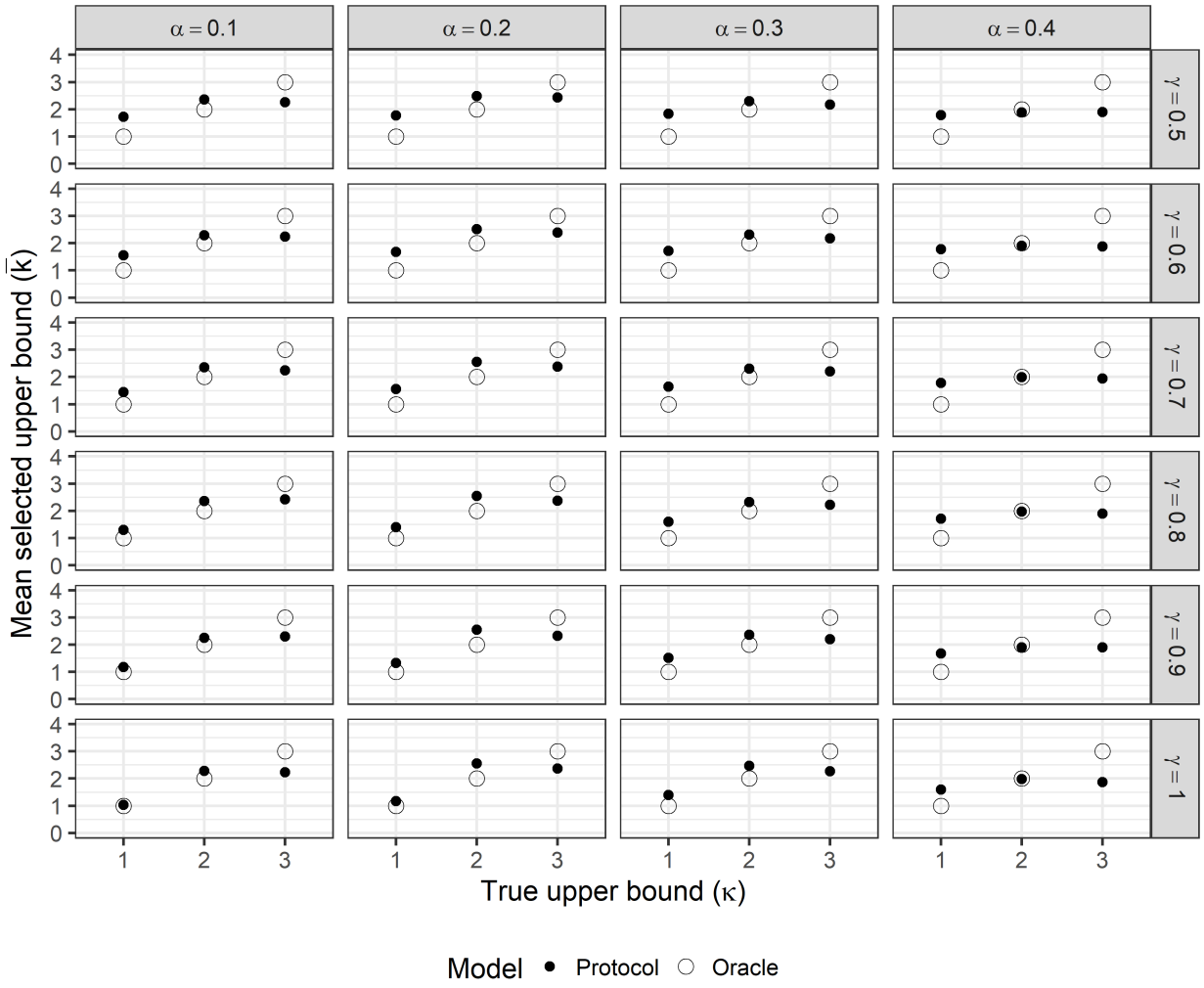

Figure A8: Mean selected upper bound for simulations with $\tau = 0.63$

B. Extended Frequentist Lasso Results

Figure B1 shows the corresponding lasso coefficients at the corresponding value of λ, which informs model selection for both the lasso and lasso + OLS. Figure B2 shows the coefficients along λ for the adaptive lasso. The blue area indicates the range between the value of λ that minimizes root mean squared error (RMSE) and λ with RMSE within one standard deviation from it. As the coefficients converge towards zero, the interpretation is to exclude them from the estimation step. See the main text for further details.

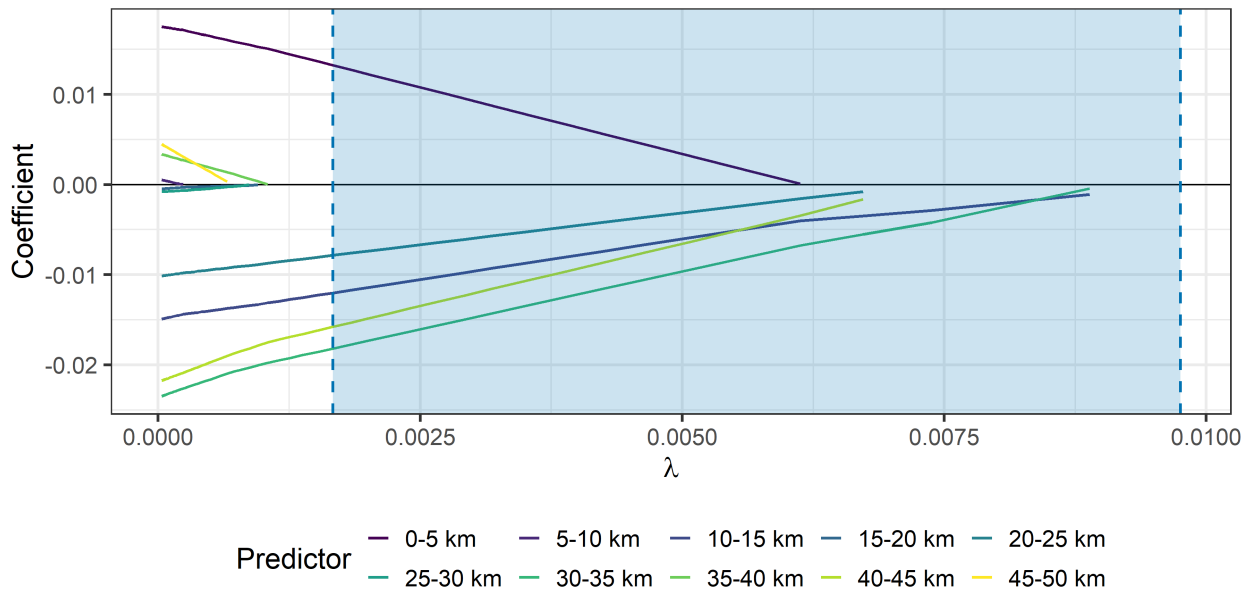

Figure B1: Lasso fit along λ

Note: Colors denote the different predictors. The blue area denotes the values of λ with RMSE within one standard deviation of the minimum, which is the leftmost vertical dashed line. As coefficients converge to zero, the interpretation is to exclude them from estimation.

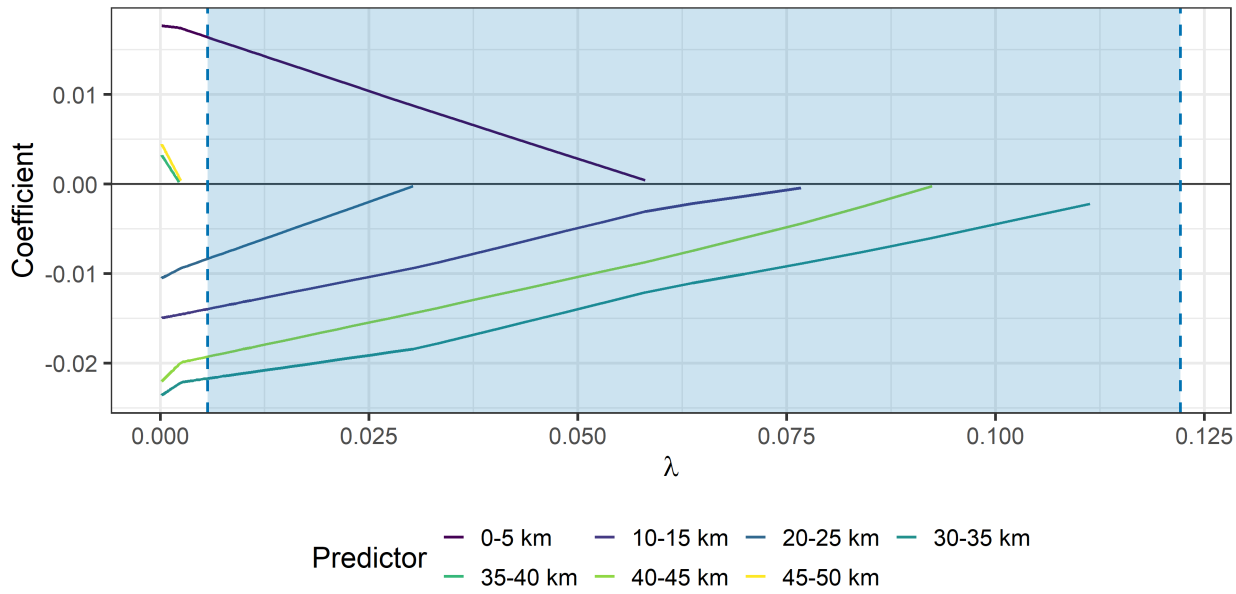

Figure B2: Adaptive lasso fit along λ

Note: Colors denote the different predictors. The blue area denotes the values of λ with RMSE within one standard deviation of the minimum, which is the leftmost vertical dashed line. As coefficients converge to zero, the interpretation is to exclude them from estimation. Predictors not displayed have a coefficient of zero throughout the values of λ.